

ARCTIC VILLAGE, AK, USA

WMO: 701945

Lat: 68.115N

Lon: 145.579W

Elev: 2092

StdP: 13.62

Time Zone: -9.00 (NAK)

Period: 95-19

WBAN: 46405

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-47.0	-43.7	-31.2	1.0	-26.3	-29.3	1.1	-21.6	19.6	5.7	16.6	3.6	0.9	40	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	15.6	72.8	55.3	70.1	54.2	66.2	52.4	57.2	68.6	55.6	66.4	54.2	63.9	7.7	240

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
52.4	63.2	59.4	51.2	60.3	59.4	48.2	53.8	58.8	25.3	68.0	24.3	66.0	23.4	63.6	62.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 17.6	15.1	12.4	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		
(5)			N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	18.4	-17.1	-9.0	-1.2	13.9	38.1	54.3	57.0	49.1	37.6	15.6	-6.7	-11.9
	DBStd	28.87	16.03	13.91	11.40	10.96	10.85	6.39	5.17	6.10	7.79	10.79	15.35	14.53
	HDD50	11983	2081	1653	1587	1083	387	31	10	89	374	1067	1701	1920
	HDD65	16996	2546	2073	2052	1533	835	323	251	492	823	1532	2151	2385
	CDD50	465	0	0	0	0	16	160	226	62	1	0	0	0
	CDD65	4	0	0	0	0	0	2	2	0	0	0	0	0
	CDH74	52	0	0	0	0	0	19	28	5	0	0	0	0
	CDH80	2	0	0	0	0	0	0	2	0	0	0	0	0
	WSAvg	4.5	2.7	2.6	3.8	4.9	5.9	6.7	6.3	6.0	5.1	3.8	2.8	3.2
	PrecAvg	10.0	0.6	0.5	0.4	0.3	0.5	1.1	1.7	1.8	1.1	0.9	0.6	0.6
(15) Precipitation	PrecMax	13.7	1.5	1.3	0.8	1.1	1.1	2.4	2.9	3.0	2.3	1.5	1.7	1.4
	PrecMin	6.1	0.2	0.1	0.1	0.1	0.2	0.4	0.7	0.5	0.2	0.2	0.0	0.2
	PrecStd	1.7	0.3	0.3	0.2	0.2	0.2	0.6	0.5	0.6	0.6	0.3	0.4	0.3
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.4	23.4	30.2	45.2	69.5	76.6	77.5	73.3	59.2	41.6	26.7	23.6
		MCWB	20.5	21.3	27.0	36.1	50.0	56.0	57.5	56.7	46.6	36.1	24.1	20.7
	2%	DB	14.2	18.6	24.9	38.6	63.3	72.5	73.1	67.0	55.2	35.7	21.3	18.2
		MCWB	13.9	17.3	21.3	31.7	48.0	54.0	55.8	54.7	44.8	32.2	19.8	17.0
	5%	DB	10.0	14.1	20.5	35.7	58.7	69.8	70.5	63.4	52.7	32.4	18.3	12.9
		MCWB	9.2	13.1	17.7	29.8	45.9	52.7	55.3	51.5	43.8	30.0	16.7	12.1
	10%	DB	6.9	9.9	16.3	31.8	54.5	65.9	67.8	60.6	48.5	29.7	14.2	8.5
		MCWB	4.7	8.8	13.9	27.5	43.7	50.9	54.0	50.1	42.1	27.9	13.4	6.9
	0.4%	WB	21.5	23.5	27.4	36.6	51.6	57.6	59.6	58.6	48.2	37.1	24.4	21.4
		MCDB	22.4	25.5	29.4	44.8	64.9	70.9	74.2	70.3	56.3	40.1	26.7	24.2
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	16.4	17.6	22.0	32.1	49.3	55.8	57.9	55.5	46.2	32.7	21.0	17.6
		MCDB	17.3	18.8	24.6	37.9	61.7	68.7	69.4	65.6	53.4	34.7	22.7	19.1
	5%	WB	11.0	13.6	18.0	29.9	46.8	54.1	56.5	53.1	44.5	30.1	17.3	13.6
		MCDB	11.1	14.7	20.7	34.5	57.4	65.9	66.9	60.7	51.1	31.9	18.8	14.6
	10%	WB	8.0	9.6	14.3	27.5	44.3	52.3	55.2	51.0	42.8	27.6	14.2	8.3
		MCDB	9.5	10.2	16.6	31.9	53.6	63.7	65.1	58.3	48.7	29.2	15.4	9.7
(35) Mean Daily Temperature Range	5% DB	MDBR	14.8	16.1	22.5	22.0	17.1	16.4	15.6	15.1	13.9	12.5	14.1	15.9
		MCDBR	16.7	15.2	24.0	22.1	21.7	20.6	20.4	19.4	20.4	13.2	15.0	18.9
		MCWBR	16.0	14.3	21.3	16.6	11.5	8.6	8.7	9.8	11.8	10.5	14.3	18.4
	5% WB	MCDBR	17.0	15.0	23.1	20.4	20.4	17.9	17.8	16.7	17.5	11.8	14.8	18.9
		MCWBR	16.5	14.2	20.6	15.5	10.9	7.9	8.3	8.9	10.6	9.8	14.3	18.4
(40) Clear-Sky Solar Irradiance	taub		0.160	0.219	0.239	0.307	0.319	0.340	0.356	0.320	0.288	0.223	0.155	N/A
	taud		1.905	2.114	2.263	2.112	2.283	2.412	2.394	2.503	2.491	2.208	1.921	N/A
	Ebn at Noon		43	192	262	264	276	273	263	264	244	188	38	0
	Edh at Noon		3	14	23	36	34	31	31	24	19	12	2	0
(44) All-Sky Solar Radiation	RadAvg		13	179	745	1485	1840	1873	1576	1147	669	244	34	0
	RadStd		1	12	36	65	140	170	109	107	74	29	4	0

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page